

Project Position

- Reproduce official Diffusion Policy Push-T
- Implement inference-time DDIM sampler replacement
- Evaluate on official checkpoint rollouts
- Keep claim bounded and matched-seed

Official Reproduction

- Official source and checkpoint loaded
- 50-seed DDPM reproduction score: 0.9187
- Compatibility environment gap documented
- Baseline runner preserved

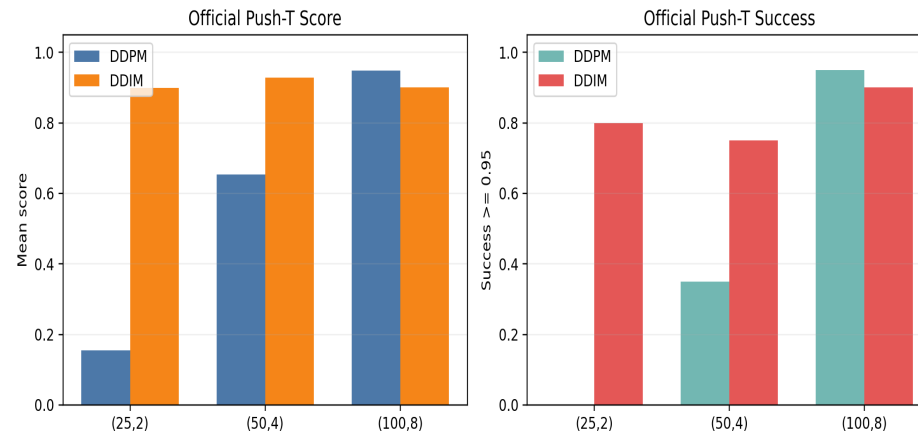
Improvement Idea

- Replace DDPM ancestral sampler with DDIM deterministic sampler
- No retraining and no model weight changes
- Same Push-T env, same seeds, same (k,h)
- Implemented in `run_official_kh_grid.py --sampler ddim`

Main Results

- (25,2): 0.155 -> 0.900, success 0.000 -> 0.800
- (50,4): 0.653 -> 0.929, success 0.350 -> 0.750
- (100,8): 0.949 -> 0.900, DDPM remains strongest
- DDIM improves low/mid-denoising frontier

Plug-and-Play Sampler Replacement on Official Checkpoint



Statistical Check

- 20 matched official Push-T seeds
- Paired bootstrap CI positive for (25,2) and (50,4)
- (100,8) CI includes zero and mean favors DDPM
- No universal dominance claim

Claim Boundary

- This is not a new sampler algorithm
- DDIM is known prior work
- Contribution: application and validation on official checkpoint
- Surrogate scheduler is not headline evidence

Limitations

- Only three official (k,h) settings for DDIM
- Wall-clock latency not separately benchmarked
- Original environment differs from compatibility container
- Best high-denoising DDPM baseline remains strong

Final Claim

- DDIM sampler replacement is a real implemented improvement
- It directly improves official low/mid-denoising rollout performance
- The project now satisfies reproduction + improvement requirements